Year	Nominal GDP (billions of dollars)	GDP Deflator (2005 = 100)
1960	543.3	19.0
1965	743.7	20.3
1970	1,075.9	24.8
1975	1,688.9	34.1
1980	2,862.5	48.3
1985	4,346.7	62.3
1990	5,979.6	72.7
1995	7,664.0	81.7
2000	10,289.7	89.0
2005	13,095.4	100.0
2010	14,958.3	110.0

Table 19.5 U.S. Nominal GDP and the GDP Deflator (Source: www.bea.gov)

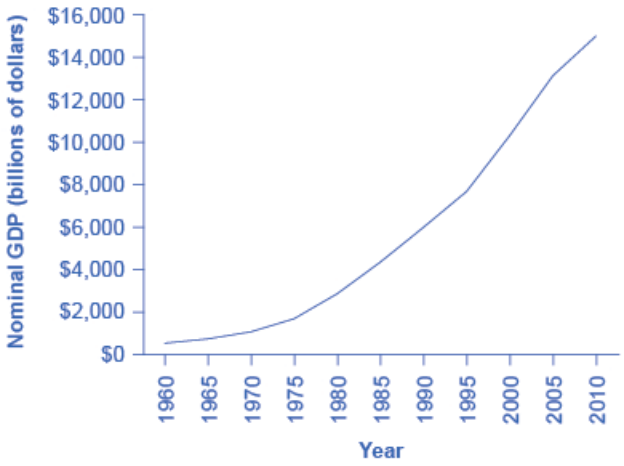


Figure 19.7 U.S. Nominal GDP, 1960–2010 Nominal GDP values have risen exponentially from 1960 through 2010, according to the BEA.

If an unwary analyst compared nominal GDP in 1960 to nominal GDP in 2010, it might appear that national output had risen by a factor of more than twenty-seven over this time (that is, GDP of $14,958 billion in 2010 divided by GDP of $543 billion in 1960 = 27.5). This conclusion would be highly misleading. Recall that we define nominal GDP as the quantity of every good or service produced multiplied by the price at which it was sold, summed up for all goods and services. In order to see how much production has actually increased, we need to extract the effects of higher prices on nominal GDP. We can easily accomplish this using the GDP deflator.

The GDP deflator is a price index measuring the average prices of all goods and services included in the economy. We explore price indices in detail and how we compute them in **Inflation**, but this definition will do in the context of this chapter. **Table 19.5** provides the GDP deflator data and **Figure 19.8** shows it graphically.

inflation, especially in the 1970s exaggerates the rise in nominal GDP.

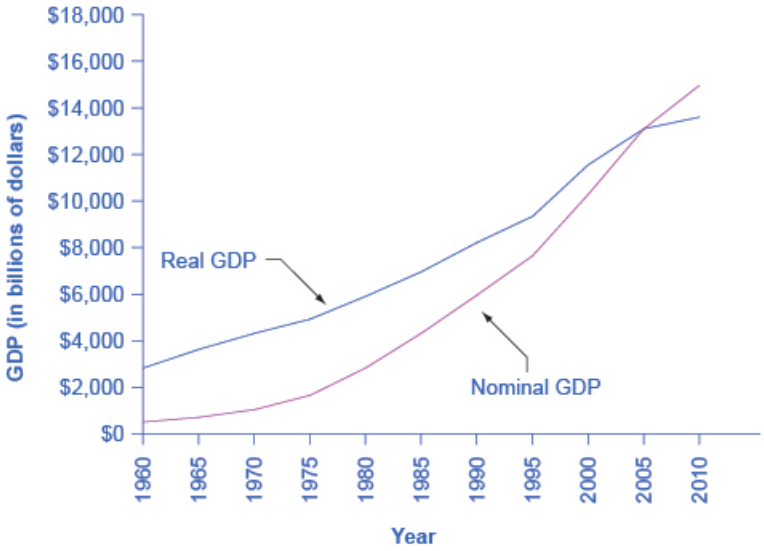


Figure 19.9 U.S. Nominal and Real GDP, 1960–2012 The red line measures U.S. GDP in nominal dollars. The black line measures U.S. GDP in real dollars, where all dollar values are converted to 2005 dollars. Since we express real GDP in 2005 dollars, the two lines cross in 2005. However, real GDP will appear higher than nominal GDP in the years before 2005, because dollars were worth less in 2005 than in previous years. Conversely, real GDP will appear lower in the years after 2005, because dollars were worth more in 2005 than in later years.

Let's return to the question that we posed originally: How much did GDP increase in real terms? What was the real GDP growth rate from 1960 to 2010? To find the real growth rate, we apply the formula for percentage change:

$$\frac{\text{2010 real GDP} - \text{1960 real GDP}}{\text{1960 real GDP}} \times 100 = \%\ \text{change}$$

$$\frac{13{,}598.5 - 2{,}859.5}{2{,}859.5} \times 100 = 376\%$$

In other words, the U.S. economy has increased real production of goods and services by nearly a factor of four since 1960. Of course, that understates the material improvement since it fails to capture improvements in the quality of products and the invention of new products.

There is a quicker way to answer this question approximately, using another math trick. Because:

$$\begin{aligned} \text{Nominal} &= \text{Price} \times \text{Quantity} \\ \%\ \text{change in Nominal} &= \%\ \text{change in Price} + \%\ \text{change in Quantity} \\ &\text{OR} \\ \%\ \text{change in Quantity} &= \%\ \text{change in Nominal} - \%\ \text{change in Price} \end{aligned}$$

Therefore, real GDP growth rate (% change in quantity) equals the growth rate in nominal GDP (% change in value) minus the inflation rate (% change in price).

Note that using this equation provides an approximation for small changes in the levels. For more accurate measures, one should use the first formula.

19.3 | Tracking Real GDP over Time

By the end of this section, you will be able to:

- Explain recessions, depressions, peaks, and troughs
- Evaluate the importance of tracking real GDP over time

When news reports indicate that "the economy grew 1.2% in the first quarter," the reports are referring to the percentage change in real GDP. By convention, governments report GDP growth is at an annualized rate: Whatever the calculated growth in real GDP was for the quarter, we multiply it by four when it is reported as if the economy were growing at that rate for a full year.

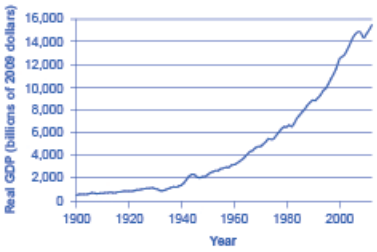


Figure 19.10 U.S. GDP, 1900–2016 Real GDP in the United States in 2016 (in 2009 dollars) was about $16.7 trillion. After adjusting to remove the effects of inflation, this represents a roughly 20-fold increase in the economy's production of goods and services since the start of the twentieth century. (Source: bea.gov)

Figure 19.10 shows the pattern of U.S. real GDP since 1900. Short term declines have regularly interrupted the generally upward long-term path of GDP. We call a significant decline in real GDP a **recession**. We call an especially lengthy and deep recession a **depression**. The severe drop in GDP that occurred during the 1930s Great Depression is clearly visible in the figure, as is the 2008–2009 Great Recession.

Real GDP is important because it is highly correlated with other measures of economic activity, like employment and unemployment. When real GDP rises, so does employment.

The most significant human problem associated with recessions (and their larger, uglier cousins, depressions) is that a slowdown in production means that firms need to lay off or fire some of their workers. Losing a job imposes painful financial and personal costs on workers, and often on their extended families as well. In addition, even those who keep their jobs are likely to find that wage raises are scanty at best—or their employers may ask them to take pay cuts.

Table 19.7 lists the pattern of recessions and expansions in the U.S. economy since 1900. We call the highest point of the economy, before the recession begins, the **peak**. Conversely, the lowest point of a recession, before a recovery begins, is the **trough**. Thus, a recession lasts from peak to trough, and an economic upswing runs from trough to peak. We call the economy's movement from peak to trough and trough to peak the **business cycle**. It is intriguing to notice that the three longest trough-to-peak expansions of the twentieth century have happened since 1960. The most recent recession started in December 2007 and ended formally in June 2009. This was the most severe recession since the 1930s Great Depression. The ongoing expansion since the June 2009 trough will also be quite long, comparatively, having already reached 90 months at the end of 2016.

Trough	Peak	Months of Contraction	Months of Expansion
December 1900	September 1902	18	21
August 1904	May 1907	23	33
June 1908	January 1910	13	19
January 1912	January 1913	24	12
December 1914	August 1918	23	44
March 1919	January 1920	7	10
July 1921	May 1923	18	22
July 1924	October 1926	14	27

Table 19.7 U.S. Business Cycles since 1900 (Source: http://www.nber.org/cycles/main.html)

21.2 | Patterns of Unemployment

By the end of this section, you will be able to:

- Explain historical patterns of unemployment in the U.S.
- Identify trends of unemployment based on demographics
- Evaluate global unemployment rates

Let's look at how unemployment rates have changed over time and how various groups of people are affected by unemployment differently.

The Historical U.S. Unemployment Rate

Figure 21.3 shows the historical pattern of U.S. unemployment since 1955.

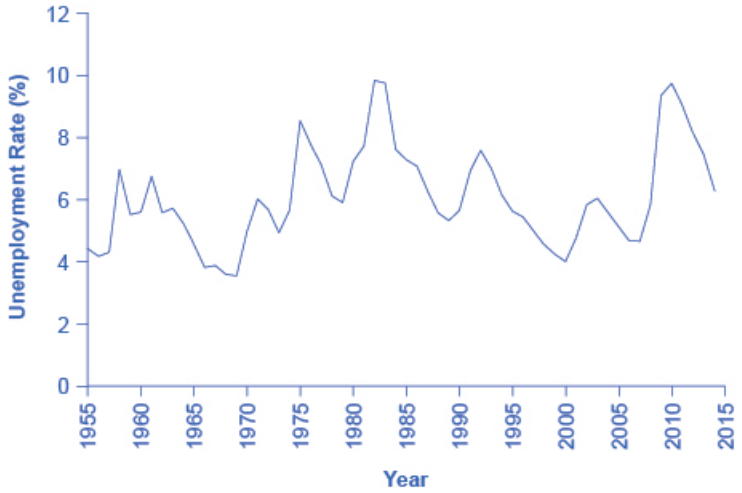


Figure 21.3 The U.S. Unemployment Rate, 1955–2015 The U.S. unemployment rate moves up and down as the economy moves in and out of recessions. However, over time, the unemployment rate seems to return to a range of 4% to 6%. There does not seem to be a long-term trend toward the rate moving generally higher or generally lower. (Source: Federal Reserve Economic Data (FRED) https://research.stlouisfed.org/fred2/series/LRUN64TTUSA156S0)

As we look at this data, several patterns stand out:

1. Unemployment rates do fluctuate over time. During the deep recessions of the early 1980s and of 2007–2009, unemployment reached roughly 10%. For comparison, during the 1930s Great Depression, the unemployment rate reached almost 25% of the labor force.
2. Unemployment rates in the late 1990s and into the mid-2000s were rather low by historical standards. The unemployment rate was below 5% from 1997 to 2000, and near 5% during almost all of 2006–2007, and 5% or slightly less from September 2015 through January 2017 (the latest date for which data are available as of this writing). The previous time unemployment had been less than 5% for three consecutive years was three

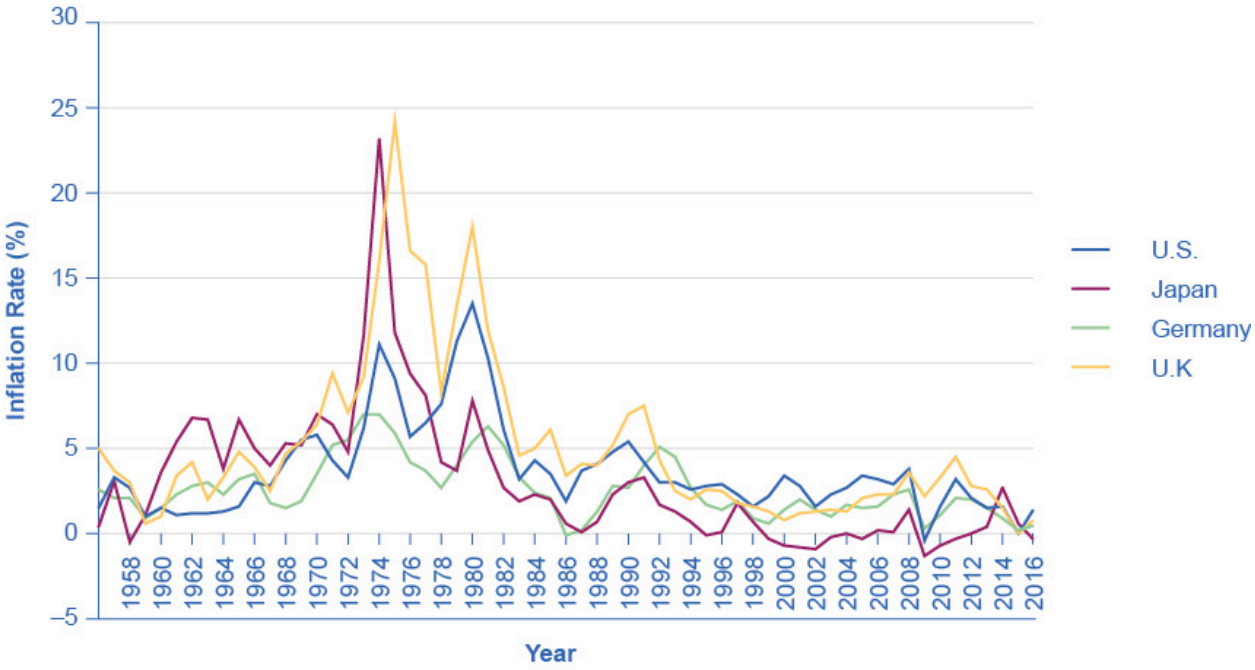


Figure 22.4 Countries with Relatively Low Inflation Rates, 1960–2016 This chart shows the annual percentage change in consumer prices compared with the previous year's consumer prices in the United States, the United Kingdom, Japan, and Germany.

Countries with controlled economies in the 1970s, like the Soviet Union and China, historically had very low rates of measured inflation—because prices were forbidden to rise by law, except for the cases where the government deemed a price increase to be due to quality improvements. However, these countries also had perpetual shortages of goods, since forbidding prices to rise acts like a price ceiling and creates a situation where quantity demanded often exceeds quantity supplied. As Russia and China made a transition toward more market-oriented economies, they also experienced outbursts of inflation, although we should regard the statistics for these economies as somewhat shakier. Inflation in China averaged about 10% per year for much of the 1980s and early 1990s, although it has dropped off since then. Russia experienced **hyperinflation**—an outburst of high inflation—of 2,500% per year in the early 1990s, although by 2006 Russia's consumer price inflation had dipped below 10% per year, as **Figure 22.5** shows. The closest the United States has ever reached hyperinflation was during the 1860–1865 Civil War, in the Confederate states.

Introduction to the Keynesian Perspective

In this chapter, you will learn about:

- Aggregate Demand in Keynesian Analysis
- The Building Blocks of Keynesian Analysis
- The Phillips Curve
- The Keynesian Perspective on Market Forces

We have learned that the level of economic activity, for example output, employment, and spending, tends to grow over time. In **The Keynesian Perspective** we learned the reasons for this trend. **The Macroeconomic Perspective** pointed out that the economy tends to cycle around the long-run trend. In other words, the economy does not always grow at its average growth rate. Sometimes economic activity grows at the trend rate, sometimes it grows more than the trend, sometimes it grows less than the trend, and sometimes it actually declines. You can see this cyclical behavior in **Figure 25.2**.

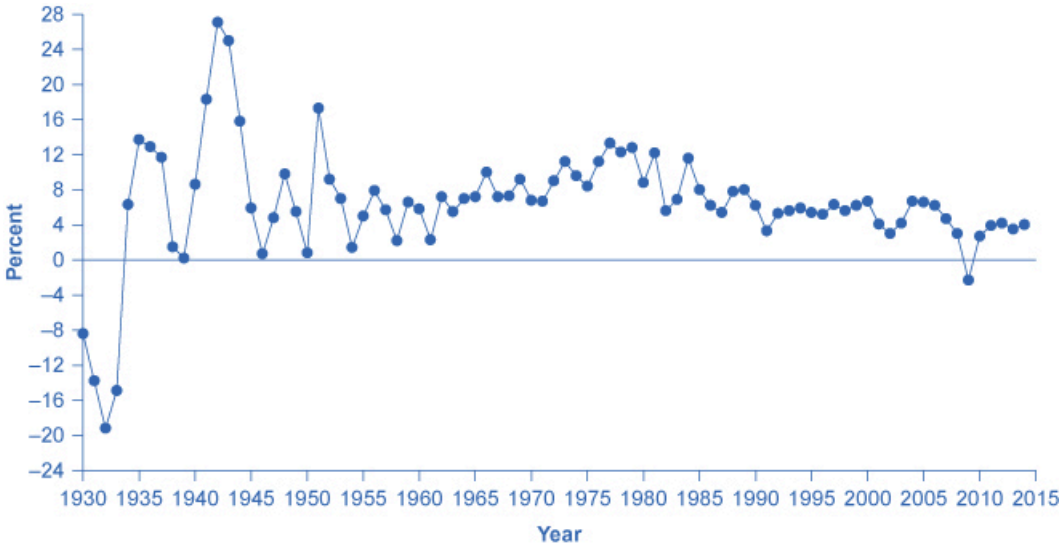


Figure 25.2 U.S. Gross Domestic Product, Percent Changes 1930–2014 The chart tracks the percent change in GDP since 1930. The magnitude of both recessions and peaks was quite large between 1930 and 1945. (Source: Bureau of Economic Analysis, "National Economic Accounts")

This empirical reality raises two important questions: How can we explain the cycles, and to what extent can we moderate them? This chapter (on the Keynesian perspective) and **The Neoclassical Perspective** explore those questions from two different points of view, building on what we learned in **The Aggregate Demand/Aggregate Supply Model**.

25.1 | Aggregate Demand in Keynesian Analysis

By the end of this section, you will be able to:

- Explain real GDP, recessionary gaps, and inflationary gaps
- Recognize the Keynesian AD/AS model
- Identify the determining factors of both consumption expenditure and investment expenditure
- Analyze the factors that determine government spending and net exports

The Keynesian perspective focuses on aggregate demand. The idea is simple: firms produce output only if they expect it to sell. Thus, while the availability of the factors of production determines a nation's potential GDP, the amount of

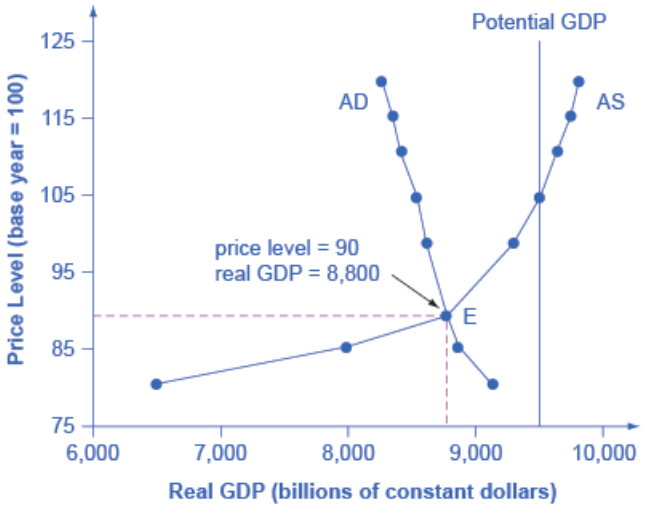


Figure 24.6 Aggregate Supply and Aggregate Demand The equilibrium, where aggregate supply (AS) equals aggregate demand (AD), occurs at a price level of 90 and an output level of 8,800.

Confusion sometimes arises between the aggregate supply and aggregate demand model and the microeconomic analysis of demand and supply in particular markets for goods, services, labor, and capital. Read the following Clear It Up feature to gain an understanding of whether AS and AD are macro or micro.

Clear It Up

Are AS and AD macro or micro?

These aggregate supply and demand models and the microeconomic analysis of demand and supply in particular markets for goods, services, labor, and capital have a superficial resemblance, but they also have many underlying differences.

For example, the vertical and horizontal axes have distinctly different meanings in macroeconomic and microeconomic diagrams. The vertical axis of a microeconomic demand and supply diagram expresses a price (or wage or rate of return) for an individual good or service. This price is implicitly relative: it is intended to be compared with the prices of other products (for example, the price of pizza relative to the price of fried chicken). In contrast, the vertical axis of an aggregate supply and aggregate demand diagram expresses the level of a price index like the Consumer Price Index or the GDP deflator—combining a wide array of prices from across the economy. The price level is absolute: it is not intended to be compared to any other prices since it is essentially the average price of all products in an economy. The horizontal axis of a microeconomic supply and demand curve measures the quantity of a particular good or service. In contrast, the horizontal axis of the aggregate demand and aggregate supply diagram measures GDP, which is the sum of all the final goods and services produced in the economy, not the quantity in a specific market.

In addition, the economic reasons for the shapes of the curves in the macroeconomic model are different from the reasons behind the shapes of the curves in microeconomic models. Demand curves for individual goods or services slope down primarily because of the existence of substitute goods, not the wealth effects, interest rate, and foreign price effects associated with aggregate demand curves. The slopes of individual supply and demand curves can have a variety of different slopes, depending on the extent to which quantity demanded and quantity supplied react to price in that specific market, but the slopes of the AS and AD curves are much the same in every diagram (although as we shall see in later chapters, short-run and long-run perspectives will emphasize different parts of the AS curve).

In short, just because the AD/AS diagram has two lines that cross, do not assume that it is the same as every other diagram where two lines cross. The intuitions and meanings of the macro and micro diagrams are only